(Lattices-based) Oblivious PRFs

Advanced Topics in ~~Cybersecurity~~Cryptography (7CCSMATC)

Martin R. Albrecht

Oblivious PRFs

(Verifiable) Oblivious PRFs

$$\begin{array}{ccccc} \text{Client} & & \text{Functionality} & & \text{Server} \\ & \xrightarrow{\quad m \quad} & & \xleftarrow{\quad k \quad} & \\ & & c = F_k(m) & & \\ & \xleftarrow{\quad c \quad} & & \xrightarrow{\quad \perp \quad} & \end{array}$$

Applications: Privacy Pass I

Privacy Pass

A privacy-enhancing protocol and browser extension.

Install:

- Home
- Protocol design
- FAQ
- Team
- Extension code
- Server code

Privacy Pass is a browser extension with the aim of making the internet more accessible.

Updates

Oct 2020: The Privacy Pass protocol is now in the process of being standardised by the IETF in the privacypass working group. All contributions are welcome! See the GitHub page for more details.

Oct 2019: Version 2.0 of the extension is now available in Chrome and Firefox!

April 2018: "Privacy Pass: Bypassing Internet Challenges Anonymously" accepted to PETS 2018. See below and in the FAQ for more details.

How?

Privacy Pass interacts with supporting websites to introduce an anonymous user-authentication mechanism. In particular, Privacy Pass is suitable for cases where a user is required to complete some proof-of-work (e.g. solving an internet challenge) to authenticate to a service. In short, the extension receives *blindly signed* 'passes' for each authentication and these passes can be used to bypass future challenge solutions using an *anonymous redemption* procedure. For example, Privacy Pass is supported by Cloudflare to enable users to redeem passes instead of having to solve CAPTCHAs to visit Cloudflare-protected websites.

Applications: Privacy Pass II

Problem:

- Tor users are having a hard time on Cloudflare protected sites
- They're constantly asked to solve CAPTCHAs to prove that they're not bots
- Want a privacy-preserving way of running reverse Turing test once and re-use later

Idea:

- Solve CAPTCHA
- Evaluate a VOPRF on a bunch of random points to produce tokens $F_k(x_i)$
- Redeem token by sending $(x_i, F_k(x_i))$

Alex Davidson, Ian Goldberg, Nick Sullivan, George Tankersley, and Filippo Valsorda. Privacy Pass: Bypassing Internet Challenges Anonymously. In: *PoPETs* 2018.3 (July 2018), pp. 164–180. DOI: `10.1515/popets-2018-0026`

Applications: OPAQUE

Problem:

- Passwords are everywhere,
- but servers know passwords, e.g.
  - phishing exploits that password are sent to server in clear
  - server breach
  - ...

Idea:

Registration Client stores $Env_C = Enc_{rwd}(sk_C, pk_S)$ on server with $rwd = F_k(pwd)$

Login run OPRF $rwd = F_k(pwd)$, client decrypts Env_C and runs key-exchange with S.

Stanislaw Jarecki, Hugo Krawczyk, and Jiayu Xu. **OPAQUE: An Asymmetric PAKE Protocol Secure Against Pre-computation Attacks.** In: *EUROCRYPT 2018, Part III*. ed. by Jesper Buus Nielsen and Vincent Rijmen. Vol. 10822. LNCS. Springer, Cham, 2018, pp. 456–486. DOI: 10.1007/978-3-319-78372-7_15

Applications: Other

- Secure keyword search[1]
- Private set intersection[2]
- Secure data de-duplication[3]
- Password-protected secret sharing[4]

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[1] Michael J. Freedman, Yuval Ishai, Benny Pinkas, and Omer Reingold. Keyword Search and Oblivious Pseudorandom Functions. In: *TCC 2005*. Ed. by Joe Kilian. Vol. 3378. LNCS. Springer, Berlin, Heidelberg, Feb. 2005, pp. 303–324. DOI: 10.1007/978-3-540-30576-7_17.

[2] Stanislaw Jarecki and Xiaomin Liu. Efficient Oblivious Pseudorandom Function with Applications to Adaptive OT and Secure Computation of Set Intersection. In: *TCC 2009*. Ed. by Omer Reingold. Vol. 5444. LNCS. Springer, Berlin, Heidelberg, Mar. 2009, pp. 577–594. DOI: 10.1007/978-3-642-00457-5_34.

[3] Sriram Keelveedhi, Mihir Bellare, and Thomas Ristenpart. DupLESS: Server-Aided Encryption for Deduplicated Storage. In: *USENIX Security 2013*. Ed. by Samuel T. King. USENIX Association, Aug. 2013, pp. 179–194. URL: https://www.usenix.org/conference/usenixsecurity13/technical-sessions/presentation/bellare.

[4] Stanislaw Jarecki, Aggelos Kiayias, Hugo Krawczyk, and Jiayu Xu. Highly-Efficient and Composable Password-Protected Secret Sharing (Or: How to Protect Your Bitcoin Wallet Online). In: *2016 IEEE European Symposium on Security and Privacy*. IEEE Computer Society Press, Mar. 2016, pp. 276–291. DOI: 10.1109/EuroSP.2016.30.

DH-Based OPRF

$$\begin{array}{ccc} \textbf{Client} & & \textbf{Server} \\ & \xrightarrow{\quad a = \mathsf{H}(x) \cdot g^r \quad} & \\ & \xleftarrow{\quad b = a^k, v = g^k \quad} & \\ \mathsf{H}(x)^k = b/v^r & & \end{array}$$

$$b/v^r = a^k/v^r = (\mathsf{H}(x) \cdot g^r)^k/(g^k)^r = \mathsf{H}(x)^k$$

Security Model

Main Reference

Nirvan Tyagi, Sofía Celi, Thomas Ristenpart, Nick Sullivan, Stefano Tessaro, and Christopher A. Wood. A Fast and Simple Partially Oblivious PRF, with Applications. In: *EUROCRYPT 2022, Part II*. ed. by Orr Dunkelman and Stefan Dziembowski. Vol. 13276. LNCS. Springer, Cham, 2022, pp. 674–705. DOI: `10.1007/978-3-031-07085-3_23`

Karla Friedrichs, Anja Lehmann, and Cavit Özbay. Game Changer: A Modular Framework for OPRF Security. In: *ASIACRYPT 2025, Part VIII*. ed. by Goichiro Hanaoka and Bo-Yin Yang. Vol. 16252. LNCS. Springer, Singapore, Dec. 2025, pp. 582–613. DOI: `10.1007/978-981-95-5125-5_19`

Syntax I

An oblivious PRF (POPRF) $\mathcal{F}$ is a tuple of PPT algorithms

$$(\mathcal{F}.\mathsf{Setup}, \mathcal{F}.\mathsf{KeyGen}, \mathcal{F}.\mathsf{Request}, \mathcal{F}.\mathsf{BlindEval}, \mathcal{F}.\mathsf{Finalise}, \mathcal{F}.\mathsf{Eval})$$

The setup and key generation algorithm generate public parameter pp and a public/secret key pair (pk, sk).

The unblinded evaluation algorithm $\mathcal{F}.\mathsf{Eval}$ is deterministic and takes as input a secret key sk, an input x and outputs a PRF evaluation z.

Syntax II

Oblivious evaluation is carried out as an interactive protocol between C and S, here presented as algorithms $\mathcal{F}$.Request, $\mathcal{F}$.BlindEval, $\mathcal{F}$.Finalise working as follows:

- First, C runs the algorithm $\mathcal{F}.\mathsf{Request}_{\mathsf{pp}}^{\mathsf{RO}}(\mathsf{pk}, x)$ taking a public key pk and a private input x. It outputs a local state *st* and a request message *req*, which is sent to the server.
- S runs $\mathcal{F}.\mathsf{BlindEval}_{\mathsf{pp}}^{\mathsf{RO}}(\mathsf{sk}, req)$ taking as input a secret key sk and the request message *req*. It produces a response message *rep* sent back to C.
- Finally, C runs $\mathcal{F}.\mathsf{Finalise}(st, rep)$ which takes the response message and its previously constructed state *st* and either outputs a PRF evaluation or $\perp$ if *rep* is rejected.

Request Privacy

Game $\mathsf{POPRIV1}^{\mathcal{A},b}_{\mathcal{F},\mathsf{RO}}(\lambda)$	Oracle $\mathsf{Run}(x_0, x_1)$
$\mathsf{pp} \leftarrow \mathcal{F}.\mathsf{Setup}(1^\lambda)$	**for** $j \in \{0,1\}$ **do**
$(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathcal{F}.\mathsf{KeyGen}^{\mathsf{RO}}_{\mathsf{pp}}(1^\lambda)$	$\quad (st_j, req_j) \leftarrow\!\!\$\ \mathcal{F}.\mathsf{Request}^{\mathsf{RO}}_{\mathsf{pp}}(\mathsf{pk}, x_j)$
$b' \leftarrow \mathcal{A}^{\mathsf{Run},\mathsf{RO}}(\mathsf{pp}, \mathsf{pk}, \mathsf{sk})$	$\quad rep_j \leftarrow\!\!\$\ \mathcal{F}.\mathsf{BlindEval}^{\mathsf{RO}}_{\mathsf{pp}}(sk, req_j)$
return b'	$\quad z_j \leftarrow\!\!\$\ \mathcal{F}.\mathsf{Finalise}^{\mathsf{RO}}_{\mathsf{pp}}(rep_j, st_j)$
	$\tau_0 \leftarrow (req_b, rep_b, z_0);\ \tau_1 \leftarrow (req_{1-b}, rep_{1-b}, z_1)$
	return (τ_0, τ_1)

We say an oblivious PRF $\mathcal{F}$ has request privacy against a honest-but-curious server if for all PPT adversary $\mathcal{A}$ the following advantage is $\mathsf{negl}(\lambda)$:

$$\mathsf{Adv}^{\mathrm{po-priv1}}_{\mathcal{F},\mathsf{S},\mathsf{RO},\mathcal{A}}(\lambda) = \left| \Pr\left[\mathsf{POPRIV1}^{\mathcal{A},1}_{\mathcal{F},\mathsf{RO}}(\lambda) \Rightarrow 1\right] - \Pr\left[\mathsf{POPRIV1}^{\mathcal{A},0}_{\mathcal{F},\mathsf{RO}}(\lambda) \Rightarrow 1\right] \right|.$$

Pseudorandomness I

We say an oblivious PRF $\mathcal{F}$ is pseudorandom if for all PPT adversaries $\mathcal{A}$, there exists a PPT simulator S such that the following advantage is $\mathrm{negl}(\lambda)$:

$$\mathsf{Adv}^{\mathrm{po-prf}}_{\mathcal{F},\mathsf{S},\mathsf{RO},\mathcal{A}}(\lambda) = \left|\Pr\left[\mathsf{POPRF}^{\mathcal{A},1}_{\mathcal{F},\mathsf{S},\mathsf{RO}}(\lambda) \Rightarrow 1\right] - \Pr\left[\mathsf{POPRF}^{\mathcal{A},0}_{\mathcal{F},\mathsf{S},\mathsf{RO}}(\lambda) \Rightarrow 1\right]\right|.$$

Game $\mathrm{POPRF}^{\mathcal{A},b}_{\mathcal{F},\mathsf{S},\mathsf{RO}}(\lambda)$

$q_s, q_t \leftarrow 0, 0$
$st_{\mathsf{Fn}} \leftarrow \mathsf{RO}_{\mathsf{Fn}}.\mathsf{Init}(1^\lambda)$
$st_{\mathsf{RO}} \leftarrow \mathsf{RO}.\mathsf{Init}(1^\lambda)$
$\mathsf{pp}_1 \leftarrow \mathcal{F}.\mathsf{Setup}(1^\lambda)$
$(st_{\mathsf{S}}, \mathsf{pk}_0, \mathsf{pp}_0) \leftarrow \mathsf{S}.\mathsf{Init}(\mathsf{pp}_1)$
$(\mathsf{sk}_1, \mathsf{pk}_1) \leftarrow \mathcal{F}.\mathsf{KeyGen}^{\mathsf{RO}}_{\mathsf{pp}_1}(1^\lambda)$
$b' \leftarrow \mathcal{A}^{\mathsf{Ev},\mathsf{Blind},\mathsf{Prim}}(\mathsf{pp}_b, \mathsf{pk}_b)$
return b'

Oracle $\mathsf{Ev}(x)$

$z_0 \leftarrow \mathsf{RO}_{\mathsf{Fn}}.\mathsf{Eval}(st_{\mathsf{Fn}}, x)$
$z_1 \leftarrow \mathcal{F}.\mathsf{Eval}^{\mathsf{RO}}_{\mathsf{pp}_1}(\mathsf{sk}_1, x)$
return z_b

Oracle $\mathsf{LimEv}(x)$

$q_s \leftarrow q_s + 1$
if $q_s \leq q_t$ **then**
  return $\mathsf{Ev}(x)$
return $\perp$

Oracle $\mathsf{Blind}(req)$

$q_t \leftarrow q_t + 1$
$(rep_0, st_{\mathsf{S}}) \leftarrow \mathsf{S}.\mathsf{BlindEval}^{\mathsf{LimEv}}(st_{\mathsf{S}}, req)$
$rep_1 \leftarrow \mathcal{F}.\mathsf{BlindEval}^{\mathsf{RO}}_{\mathsf{pp}_1}(\mathsf{sk}_1, req)$
return rep_b

Oracle $\mathsf{Prim}(x)$

$(h_0, st_{\mathsf{S}}) \leftarrow \mathsf{S}.\mathsf{Eval}^{\mathsf{LimEv}}(st_{\mathsf{S}}, x)$
$h_1 \leftarrow\!\!\$\ \mathsf{RO}.\mathsf{Eval}(st_{\mathsf{RO}}, x)$
return h_b

Pseudorandomness III

The intuition of this definition is that it requires the simulator to explain a random output (defined via $\mathsf{RO}_{\mathsf{Fn}}$) as an evaluation point of the PRF.

- The simulator provides its own public key and public parameters, but it gets at most one query to $\mathsf{RO}_{\mathsf{Fn}}()$ per BlindEval query that it has to simulate.
- The simulator queries $\mathsf{RO}_{\mathsf{Fn}}$ through calls to LimitEval, where the check $q_s \leq q_t$ enforces the number of queries per BlindEval query.
- This implies that BlindEval and Eval queries essentially leak nothing beyond a single evaluation to the client.
- Moreover, the simulator is restricted in that the LimEv oracle will error if more queries are made to it than the number of Blind queries at any point in the game.

LWE and DH

On the one hand, on the other hand

Bottom of the Stack:

- Kyber (KEM)
- Dilithium (Signature)
- Falcon (Signature)

Top of the Stack:

- FHE [Gen09; BGV12; CGGI20; GSW13]
- iO [GGHRSW13; BDGM20]
- FE [SW05; BSW11]

Somewhat efficient

RSA 2048

Key generation	≈ 130,000,000 cycles
Encapsulation	≈ 20,000 cycles
Decapsulation	≈ 2,700,000 cycles
Ciphertext	256 bytes
Public key	256 bytes

https://bench.cr.yp.to/results-kem.html

Curve25519

Key generation	≈ 60,000 cycles
Key agreement	≈ 160,000 cycles
Public key	32 bytes
Key Share	32 bytes

https://eprint.iacr.org/2015/943

Kyber-768

Key generation	≈ 38,000 cycles
Encapsulation	≈ 49,000 cycles
Decapsulation	≈ 39,000 cycles
Ciphertext	1,088 bytes
Public key	1,184 bytes

https://bench.cr.yp.to/results-kem.html

The Learning with Errors Problem (LWE)

Given $(\mathbf{A}, \mathbf{c})$ with $\mathbf{c} \in \mathbb{Z}_q^m$, $\mathbf{A} \in \mathbb{Z}_q^{m \times n}$, $\mathbf{s} \in \mathbb{Z}_q^n$ and **small $\mathbf{e} \in \mathbb{Z}^m$** is

$$\begin{pmatrix} \\ \\ \mathbf{c} \\ \\ \\ \end{pmatrix} = \begin{pmatrix} \leftarrow & n & \rightarrow \\ & & \\ & \mathbf{A} & \\ & & \\ & & \end{pmatrix} \times \begin{pmatrix} \mathbf{s} \end{pmatrix} + \begin{pmatrix} \\ \\ \mathbf{e} \\ \\ \\ \end{pmatrix}$$

or $\mathbf{c} \leftarrow_{\$} \mathcal{U}\left(\mathbb{Z}_q^m\right)$.

LWE

$$\begin{pmatrix} c_0 \\ c_1 \\ c_2 \\ c_3 \\ c_4 \\ c_5 \\ c_6 \\ c_7 \end{pmatrix} = \begin{pmatrix} a_{0,0} & a_{0,1} & a_{0,2} & a_{0,3} & a_{0,4} & a_{0,5} & a_{0,6} & a_{0,7} \\ a_{1,0} & a_{1,1} & a_{1,2} & a_{1,3} & a_{1,4} & a_{1,5} & a_{1,6} & a_{1,7} \\ a_{2,0} & a_{2,1} & a_{2,2} & a_{2,3} & a_{2,4} & a_{2,5} & a_{2,6} & a_{2,7} \\ a_{3,0} & a_{3,1} & a_{3,2} & a_{3,3} & a_{3,4} & a_{3,5} & a_{3,6} & a_{3,7} \\ a_{4,0} & a_{4,1} & a_{4,2} & a_{4,3} & a_{4,4} & a_{4,5} & a_{4,6} & a_{4,7} \\ a_{5,0} & a_{5,1} & a_{5,2} & a_{5,3} & a_{5,4} & a_{5,5} & a_{5,6} & a_{5,7} \\ a_{6,0} & a_{6,1} & a_{6,2} & a_{6,3} & a_{6,4} & a_{6,5} & a_{6,6} & a_{6,7} \\ a_{7,0} & a_{7,1} & a_{7,2} & a_{7,3} & a_{7,4} & a_{7,5} & a_{7,6} & a_{7,7} \end{pmatrix} \cdot \begin{pmatrix} s_0 \\ s_1 \\ s_2 \\ s_3 \\ s_4 \\ s_5 \\ s_6 \\ s_7 \end{pmatrix} + \begin{pmatrix} e_0 \\ e_1 \\ e_2 \\ e_3 \\ e_4 \\ e_5 \\ e_6 \\ e_7 \end{pmatrix}$$

Performance

Storage: $\mathcal{O}(n^2)$; Computation $\mathcal{O}(n^2)$

Ring-LWE/Polynomial-LWE

$$\begin{pmatrix} c_0 \\ c_1 \\ c_2 \\ c_3 \\ c_4 \\ c_5 \\ c_6 \\ c_7 \end{pmatrix} = \begin{pmatrix} a_0 & -a_7 & -a_6 & -a_5 & -a_4 & -a_3 & -a_2 & -a_1 \\ a_1 & a_0 & -a_7 & -a_6 & -a_5 & -a_4 & -a_3 & -a_2 \\ a_2 & a_1 & a_0 & -a_7 & -a_6 & -a_5 & -a_4 & -a_3 \\ a_3 & a_2 & a_1 & a_0 & -a_7 & -a_6 & -a_5 & -a_4 \\ a_4 & a_3 & a_2 & a_1 & a_0 & -a_7 & -a_6 & -a_5 \\ a_5 & a_4 & a_3 & a_2 & a_1 & a_0 & -a_7 & -a_6 \\ a_6 & a_5 & a_4 & a_3 & a_2 & a_1 & a_0 & -a_7 \\ a_7 & a_6 & a_5 & a_4 & a_3 & a_2 & a_1 & a_0 \end{pmatrix} \cdot \begin{pmatrix} s_0 \\ s_1 \\ s_2 \\ s_3 \\ s_4 \\ s_5 \\ s_6 \\ s_7 \end{pmatrix} + \begin{pmatrix} e_0 \\ e_1 \\ e_2 \\ e_3 \\ e_4 \\ e_5 \\ e_6 \\ e_7 \end{pmatrix}$$

Performance (n is a power of two)

Storage: $\mathcal{O}(n)$; Computation $\mathcal{O}(n \log n)$

Ring-LWE/Polynomial-LWE

$$\sum_{i=0}^{n-1} c_i \cdot X^i = \left(\sum_{i=0}^{n-1} a_i \cdot X^i\right) \cdot \left(\sum_{i=0}^{n-1} s_i \cdot X^i\right) + \sum_{i=0}^{8} e_i \cdot X^i \bmod X^n + 1$$

$$c(X) = a(X) \cdot s(X) + e(X) \bmod \phi(X)$$

We write $\mathcal{R} := \mathbb{Z}[X]/\phi(X)$ and $\mathcal{R}_q := \mathbb{Z}_q[X]/\phi(X)$.

Damien Stehlé, Ron Steinfeld, Keisuke Tanaka, and Keita Xagawa. Efficient Public Key Encryption Based on Ideal Lattices. In: *ASIACRYPT 2009*. Ed. by Mitsuru Matsui. Vol. 5912. LNCS. Springer, Berlin, Heidelberg, Dec. 2009, pp. 617–635. DOI: `10.1007/978-3-642-10366-7_36`

Vadim Lyubashevsky, Chris Peikert, and Oded Regev. On Ideal Lattices and Learning with Errors over Rings. In: *EUROCRYPT 2010*. Ed. by Henri Gilbert. Vol. 6110. LNCS. Springer, Berlin, Heidelberg, 2010, pp. 1–23. DOI: `10.1007/978-3-642-13190-5_1`

Convention

- I am going to use the Ring-LWE formulation

$$c_i(X) = a_i(X) \cdot s(X) + e_i(X)$$

  Thus, each sample corresponds to "n LWE samples"
- I will suppress the "(X)" in "$a(X)$" etc.
- I will assume s is "small" and that the product of two "small" things is "small".
- I will write e_i to emphasise that e_i is small.

TL;DR: I will write

$$c_i = a_i \cdot s + e_i$$

DH to Ring-LWE Dictionary

DH Land	Ring-LWE Land
g	a
g^x	$a \cdot s + e$
$g^x \cdot g^y = g^{x+y}$	$(a \cdot s + e_0) + (a \cdot t + e_1) = a \cdot (s + t) + e'$
$(g^a)^b = (g^b)^a$	$(a \cdot s + e) \cdot t = (a \cdot s \cdot t + e \cdot t)$ $\approx a \cdot s \cdot t \approx (a \cdot t + e) \cdot s$
(g, g^a, g^b, g^{ab}) $\approx_c (g, g^a, g^b, u)$	$(a,\ a \cdot s + e,\ a \cdot t + d,\ a \cdot s \cdot t + e')$ $\approx_c (a,\ a \cdot s + e,\ a \cdot t + d,\ u)$

ElGamal & LPR10

ElGamal

KeyGen $h = g^s$

Encrypt $d_0,\ d_1 = (g^v,\ m \cdot h^v)$ for some random v

Decrypt $d_1/d_0^s = m \cdot (g^s)^v/(g^v)^s = m$

[LPR10]

KeyGen $c = a \cdot s + e$

Encrypt $d_0,\ d_1 = v \cdot a + e',\ v \cdot c + e'' + \left\lfloor \frac{q}{2} \right\rfloor \cdot m$

Decrypt

$$\left\lfloor \frac{2}{q} \cdot (d_1 - d_0 \cdot s) \right\rceil = \left\lfloor \frac{2}{q} \cdot \left(v \cdot (a \cdot s + e) + e'' + \left\lfloor \frac{q}{2} \right\rfloor \cdot m - (v \cdot a + e') \cdot s \right) \right\rceil$$

$$= \left\lfloor \frac{2}{q} \cdot \left(v \cdot e + e'' + \left\lfloor \frac{q}{2} \right\rfloor \cdot m - e' \cdot s \right) \right\rceil = m$$

Follow the Blueprint

DH-Based OPRF

Client		Server
	$\xrightarrow{c_x := \mathsf{H}(x) \cdot g^r}$	
	$\xleftarrow{d_x := c_x^k,\, c = g^k}$	
$d_x / c^r = \mathsf{H}(x)^k$		

$$d_x / c^r = c_x^k / c^r = (\mathsf{H}(x) \cdot g^r)^k / (g^k)^r = \mathsf{H}(x)^k$$

"Just take Logs"

Client | Server

$$c_x := \mathsf{H}(x) + a \cdot r + e \quad \longrightarrow$$

$$\longleftarrow \quad d_x := c_x \cdot k + e', c := a \cdot k + e''$$

$$\left\lfloor \frac{p}{q} \cdot d_x - c \cdot r \right\rceil$$

$$\begin{aligned} d_x - c \cdot r &= (\mathsf{H}(x) + a \cdot r + e) \cdot k + e' - (a \cdot k + e'') \cdot r \\ &= \mathsf{H}(x) \cdot k + a \cdot r \cdot k + e \cdot k + e' - a \cdot k \cdot r - e'' \cdot r \\ &= \mathsf{H}(x) \cdot k + e \cdot k + e' - e'' \cdot r \\ &\approx \mathsf{H}(x) \cdot k \end{aligned}$$

The Trouble

Trapdoor Friendly It is not safe to output $c_x \cdot k + e$ for some arbitrary c_x

Noise Leakage "$\approx$" glosses over $e \cdot k + e' - e'' \cdot r$ which depends on k

Noise Growth "$\approx$" is not $=$, how do we arrive at the same value?

Trapdoor Friendly

The Problem

The server has to output $c_x \cdot k + e'$ for some $c_x \stackrel{?}{=} \mathrm{H}(x) + a \cdot r + e$. This may not be safe.

Point Validation

Validation of Elliptic Curve Public Keys

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Abstract. We present practical and realistic attacks on some standardized elliptic curve key establishment and public-key encryption protocols that are effective if the receiver of an elliptic curve point does not check that the point lies on the appropriate elliptic curve. The attacks combine ideas from the small subgroup attack of Lim and Lee, and the differential fault attack of Biehl, Meyer and Müller. Although the ideas behind the attacks are quite elementary, and there are simple countermeasures known, the attacks can have drastic consequences if these countermeasures are not taken by implementors of the protocols. We illustrate the effectiveness of such attacks on a key agreement protocol recently proposed for the IEEE 802.15 Wireless Personal Area Network (WPAN) standard.

Point Validation

Validation of Elliptic Curve Public Keys

Prime and Prejudice: Primality Testing Under Adversarial Conditions

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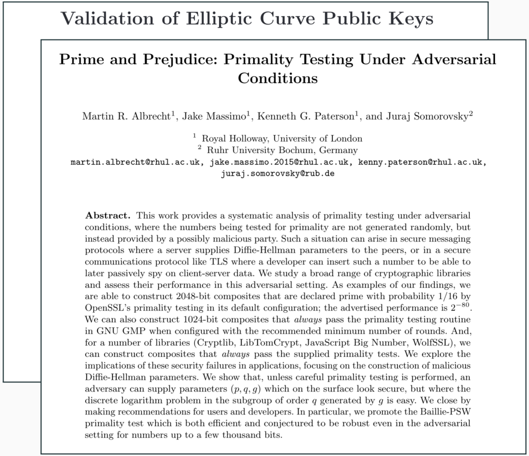
Abstract. This work provides a systematic analysis of primality testing under adversarial conditions, where the numbers being tested for primality are not generated randomly, but instead provided by a possibly malicious party. Such a situation can arise in secure messaging protocols where a server supplies Diffie-Hellman parameters to the peers, or in a secure communications protocol like TLS where a developer can insert such a number to be able to later passively spy on client-server data. We study a broad range of cryptographic libraries and assess their performance in this adversarial setting. As examples of our findings, we are able to construct 2048-bit composites that are declared prime with probability 1/16 by OpenSSL's primality testing in its default configuration; the advertised performance is 2^{-80}. We can also construct 1024-bit composites that *always* pass the primality testing routine in GNU GMP when configured with the recommended minimum number of rounds. And, for a number of libraries (Cryptlib, LibTomCrypt, JavaScript Big Number, WolfSSL), we can construct composites that *always* pass the supplied primality tests. We explore the implications of these security failures in applications, focusing on the construction of malicious Diffie-Hellman parameters. We show that, unless careful primality testing is performed, an adversary can supply parameters (p, q, g) which on the surface look secure, but where the discrete logarithm problem in the subgroup of order q generated by g is easy. We close by making recommendations for users and developers. In particular, we promote the Baillie-PSW primality test which is both efficient and conjectured to be robust even in the adversarial setting for numbers up to a few thousand bits.

Point Validation



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Abstract. This work provides a systematic analysis of primality testing under adversarial conditions, where the numbers being tested for primality are not generated randomly, but instead provided by a possibly malicious party. Such a situation can arise in secure messaging protocols where a server supplies Diffie-Hellman parameters to the peers, or in a secure communications protocol like TLS where a developer can insert such a number to be able to later passively spy on client-server data. We study a broad range of cryptographic libraries and assess their performance in this adversarial setting. As examples of our findings, we are able to construct 2048-bit composites that are declared prime with probability 1/16 by OpenSSL's primality testing in its default configuration; the advertised performance is 2^{-80}. We can also construct 1024-bit composites that *always* pass the primality testing routine in GNU GMP when configured with the recommended minimum number of rounds. And, for a number of libraries (Cryptlib, LibTomCrypt, JavaScript Big Number, WolfSSL), we can construct composites that *always* pass the supplied primality tests. We explore the implications of these security failures in applications, focusing on the construction of malicious Diffie-Hellman parameters. We show that, unless careful primality testing is performed, an adversary can supply parameters (p, q, g) which on the surface look secure, but where the discrete logarithm problem in the subgroup of order q generated by g is easy. We close by making recommendations for users and developers. In particular, we promote the Baillie-PSW primality test which is both efficient and conjectured to be robust even in the adversarial setting for numbers up to a few thousand bits.

- However, (likely) no "point validation" for LWE by the NTRU assumption [HPS96]:

$$f, g \leftarrow\!\$\; \mathcal{R}^2 : h := f/g \approx_c \mathcal{U}(\mathcal{R}_q)$$

- Attack[a]:
  1. Sample $f, g \leftarrow\!\$\; \mathcal{R}^2$ and set $\Delta := \lceil \sqrt{q} \rceil$
  2. Submit $a := \Delta \cdot f/g$
  3. Receive $c := a \cdot k + e$ and compute

$$g \cdot c = g \cdot (\Delta \cdot f/g \cdot k + e)$$
$$g \cdot c = \Delta \cdot f \cdot k + g \cdot e$$
$$\equiv g \cdot e \bmod \Delta$$

[a] assuming f, g, k, e are sufficiently small

Workaround

- The client proves in zero-knowledge that c_x is well-formed: $c_x := \mathsf{H}(x) + a \cdot r + e$
- This means the client needs to prove the evaluation of $\mathsf{H}(x)$
- This is sound, we do not need to treat $\mathsf{H}(x)$ as a Random Oracle
- This is expensive in terms of bandwidth and computation cost
    - [ADDS21]: $\approx$ 128GB per evaluation using [YAZXYW19]
    - [AG24]: $\approx$ 63kB per evaluation using [BS23]

An Aside

This NTRU “attack” can be used constructively to make proof systems online extractable (e.g. [ADDG24])

Noise Leakage

The client learns

$$e \cdot k + e' - e'' \cdot r$$

where it chooses e and r.

The Attack

Write $\mathbf{a} := (e, -r)$ and $\mathbf{s} := (k, e'')$, then we can rewrite

$$e \cdot k + e' - e'' \cdot r$$

as $\mathbf{a} \cdot \mathbf{s} + e''$ which is essentially an instance of "LWE without modular reduction" [BDEFT18] which is easy.[5]

[5]The word "essentially" does a lot of work here. That is, this is a simplification because e'' changes in each invocation and the attacks from [BDEFT18] do not apply as is.

Solutions

Statistical Noise Drowning $\|e'\| \geq \lambda^{\omega(1)} \cdot \|e \cdot k - e'' \cdot r\|$ [ADDS21]

Rényi Noise Drowning $\|e'\| \geq \mathrm{poly}(\lambda) \cdot \sqrt{Q} \cdot \|e \cdot k - e'' \cdot r\|$ [AG24]

- Q is the number of queries
- must play a search game instead of a distinguishing game (use ROM)

Computational Assumption $\|e'\| \geq \mathrm{poly}(\lambda) \cdot \sqrt{Q} \cdot \|e \cdot k - e'' \cdot r\|$ [ESTX24]

- similar to Hint-(M)LWE

Cost

Since we require $q > \|e'\|$ we have that $q/\|e\|$ – the "signal to noise ratio" of the underlying RLWE samples – is quite big. A big signal to noise ratio makes decoding – i.e. solving LWE – easier. This requires us to use larger secret dimensions n to compensate. Bandwidth cost is essentially $n \log q$.

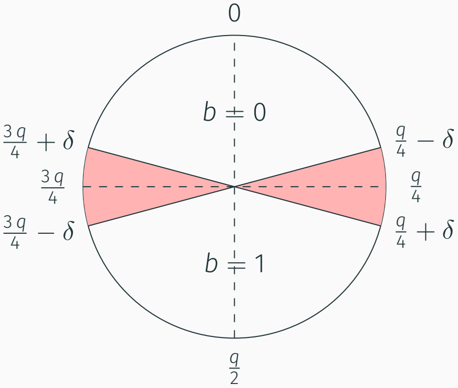
Noise Growth

The Problem

$$H(x) \cdot k + e \cdot k + e' - e'' \cdot r \approx H(x) \cdot k$$

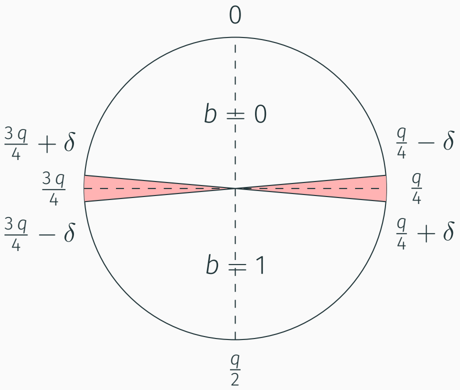
$$\left\lceil \frac{2}{q} \cdot \left(H(x) \cdot k + e \cdot k + e' - e'' \cdot r \right) \right\rfloor \stackrel{?}{=} \left\lceil \frac{2}{q} \cdot \left(H(x) \cdot k \right) \right\rfloor$$

Rounding



0
$b = 0$
$\frac{3q}{4} + \delta$
$\frac{q}{4} - \delta$
$\frac{3q}{4}$
$\frac{q}{4}$
$\frac{3q}{4} - \delta$
$\frac{q}{4} + \delta$
$b = 1$
$\frac{q}{2}$

Make $q > \mathrm{poly}(\lambda)$ such that the red area is negligibly small.

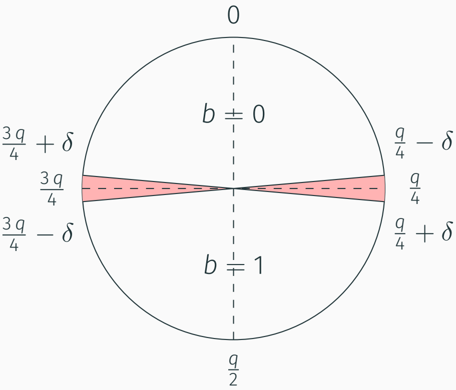


Solution Attempt

Make $q > \text{poly}(\lambda)$ such that the red area is negligibly small.

Malicious Servers

This argument works on average but does not work against adversaries that somehow pick k s.t. $\mathsf{H}(x) \cdot k$ lands in the red area with high probability for some x.



Solution

- Plant a hard SIS instance in each coefficient: 1D-SIS [ADDS21]

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  - Yes, seriously!
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- Plant a hard SIS instance in each coefficient: 1D-SIS [ADDS21]
    - Yes, seriously!
    - This requires $q \gg 2^\lambda$
- Change the PRF output to $\mathrm{H}(x) \cdot k + \mathrm{H}_2(x, c_0)$ [AG24]
    - $\mathrm{H}_2()$ is some Random Oracle that randomly shifts $\mathrm{H}(x) \cdot k$
    - $c_0 := a_0 \cdot k + e_0'$ is a commitment to k
    - The trick is from [GKQMS24]
    - $q \gg \mathrm{poly}(\lambda)$ is sufficient

Stuck with a super-polynomial q

Big "signal-to-noise" ratio, forcing us to increase n, as above.

Related: Can't Just Do It™ — NIKE

NIKE enables Alice and Bob, who know each others' public keys, to agree on shared key without requiring any interaction [DH76]

- Deployed in WireGuard [HNSWZ20] and static DH is also used in e.g. Google's QUIC.
- For lattices there are significant barriers [GKRS20].
- Stark contrast to interactive key-exchanges or plain public-key encryption
  1. We send along some "hints" that allow to handle the noise
  2. secrets are not re-used, allowing us to avoid expensive "well-formedness" proofs
- [GKQMS24] is an instantiation that essentially accepts the super-polynomial modulus

Wrapping Up

Realisations of this Blueprint

Work	Model	1-time Offline	Online	Q
[ADDS21]	H-H	–	2MB	
[ADDS21]	M-M	–	128GB	
[AG24]	M-M	114kB	198kB	2^{32}
[ESTX24]	M-M	20kB	159kB	2^{32}

H: semi-honest, M: malicious

Other Round-Optimal Alternatives w/o Trusted Setup

Work	Assumption	Model	1-time Offline	Online
ADDS21	(R)LWE+SIS	H-H	-	2MB
ADDS21	(R)LWE+SIS	M-M	-	128GB
AG24	(R)LWE+SIS	M-M	114kB	198kB
ADDG23	mod(2,3)+lattices	M-H	2.5MB	10KB
ADDG23	mod(2,3)+lattices	M-M	2.5MB	160KB
ESTX24	iMLWER-RU+MLE+SIS	M-M	20kB	159KB
APRR24	mod(2,3)	M-H, pp	4.75B	114.5B
FOO23	AES+Garbled Circuits	H-H	-	6.79MB
Basso24	Higher-Dimensional isogenies	M-M	-	28.9kB
HHM+23	Isogenies F_p + lattices + HE OT	H-H	-	640kB
dSP23	Isogenies F_p	M-H, pp	68.4 kB	384B
dSP23	Isogenies F_p	M-H, pp	-	16.38kB

adapted from `https://heimberger.xyz/oprfs.html`

Thank You

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